



ACME ACADEMY

India's Most Trustable MCA Entrance Academy

NIMCET 2021

Mathematics:

1. If $|k| = 5$ and $0^\circ \leq \theta \leq 360^\circ$, then the number of distinct solutions of $3 \cos \theta + 4 \sin \theta = k$ is
(A) 0 (B) 1
(C) 2 (D) infinite
2. The area of the region bounded by the X -axis and the curves defined by $y = \tan x$, $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$ and $y = \cot x$, $\frac{\pi}{6} \leq x \leq \frac{3\pi}{2}$ is
(A) $-\frac{1}{2} \log 2$ (B) $\frac{1}{2} \log 2$
(C) $\log 2$ (D) None of these
3. If $\log(1 - x + x^2) = a_1x + a_2x^2 + \dots$. Then, $a_3 + a_6 + a_9 + \dots$ is equal to
(A) $\log 2$ (B) $\frac{2}{3} \log 2$
(C) $\frac{1}{3} \log 2$ (D) $2 \log 2$
4. $\lim_{x \rightarrow \infty} \left(\frac{x+7}{x+2}\right)^{x+5}$ is equal to
(A) e^5 (B) e^{-5}
(C) e^2 (D) e^{-2}
5. The number of common tangents to the circle $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is
(A) 0 (B) 1
(C) 3 (D) 4
6. In a ΔABC , if $\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} = k$, then k is always
(A) > 2 (B) ≥ 1
(C) $= 2$ (D) $= 1$
7. Let $a = 2\hat{i} + \hat{j} + 2\hat{k}$, $b = \hat{i} - \hat{j} + 2\hat{k}$ and $c = \hat{i} + \hat{j} - 2\hat{k}$ are three vectors. Then, a vector in the plane of a and c whose projection on b is of magnitude $\frac{1}{\sqrt{6}}$ is
(A) $3\hat{i} - 2\hat{j}$ (B) $3\hat{i} + 2\hat{j}$
(C) $2\hat{i} + 3\hat{j} - \hat{k}$ (D) $3\hat{i} + 2\hat{j} + \hat{k}$
8. For what value of p , the polynomial $x^4 - 3x^3 + 2px^2 - 6$ is exactly divisible by $(x - 1)$?
(A) 2 (B) 4
(C) 6 (D) 8
9. If $F(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then $F(\theta)F(\alpha)$ is equal to
(A) $F(\theta\alpha)$ (B) $F\left(\frac{\theta}{\alpha}\right)$
(C) $F(\theta + \alpha)$ (D) $F(\theta - \alpha)$
10. If $\frac{n!}{2!(n-2)!}$ and $\frac{n!}{4!(n-4)!}$ are in the ratio 2 : 1, then the value of n is
(A) 0 (B) 2
(C) 4 (D) 5
11. The locus of the point of intersection of tangents to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which meet right angles is
(A) a circle (B) a parabola
(C) an ellipse (D) a hyperbola
12. If the position vector of A and B relative to O be $\hat{i} - 4\hat{j} + 3\hat{k}$ and $-\hat{i} + 2\hat{j} - \hat{k}$ respectively, then the median through O of ΔABC is
(A) $-2\hat{i} + 2\hat{k}$ (B) $-\hat{j} + \hat{k}$
(C) $-\hat{i} - \hat{j} + \hat{k}$ (D) $-\hat{i} - \hat{j} - \hat{k}$
13. The general value of θ , satisfying the equation, $\sin \theta = -\frac{1}{2}$, $\tan \theta = \frac{1}{\sqrt{3}}$ is
(A) $n\pi + \frac{\pi}{6}$, $n \in I$ (B) $n\pi + (-1)^n \left(\frac{7\pi}{6}\right)$, $n \in I$
(C) $2n\pi \pm \frac{7\pi}{6}$, $n \in I$ (D) $2n\pi + \frac{11\pi}{6}$, $n \in I$
14. The area of the triangle formed by the vertices whose position vectors are $3\hat{i} + \hat{j}$, $5\hat{i} + 2\hat{j}$, $\hat{i} - 2\hat{j} + 3\hat{k}$ is





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

(A) $\sqrt{21}$ sq. units

(B) $\sqrt{23}$ sq. units

(C) $\sqrt{33}$ sq. units

(D) $\sqrt{29}$ sq. units

15. The standard deviation of 20 number is 30. If each of the numbers is increased by 4, then the new standard deviation will be

(a) 24

(B) 34

(C) 30

(D) 20

16. The function $f(x) = \frac{x}{1+x \tan x}$, $0 \leq x \leq \frac{\pi}{2}$ is maximum when

(A) $x = \sec x$

(B) $x = \tan x$

(C) $x = \cos x$

(D) None of the above

17. If $f: R \rightarrow R$ is defined by

$$f(x) = \begin{cases} \frac{x+2}{x^2+3x+2}, & \text{if } x \in R - \{-1, -2\} \\ -1, & \text{if } x = -2 \\ 0, & \text{if } x = -1 \end{cases}$$

Then, $f(x)$ is continuous on the set

(A) R

(B) $R - \{-2\}$

(C) $R - \{-1\}$

(D) $R - \{-1, -2\}$

18. A polygon has 44 diagonals, the number of sides are

(A) 9

(B) 10

(C) 11

(D) 12

19. The probability of occurrence of two events E and F are 0.25 and 0.50, respectively. The probability of their simultaneously occurrence is 0.14. The probability that neither E nor F occur is

(A) 0.61

(B) 0.11

(C) 0.39

(D) 0.89

20. If $y = \tan^{-1} \left(\frac{3x-x^3}{1-3x^3} \right)$, $-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$, then $\frac{dy}{dx}$ is

(A) $-\frac{1}{1+x^2}$

(B) $\frac{3}{1+x^2}$

(C) $\frac{3}{\sqrt{1+x^2}}$

(D) $\frac{1}{\sqrt{1+x^2}}$

21. The value of $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$ is equal to

(A) 5

(B) 3

(C) 4

(D) 6

22. In a triangle, if the sum of two sides is x and their product is y such that $(x+z)(x-z) = y$, where z is the third side of the triangle, then triangle is

(A) equilateral

(B) right angled

(C) isosceles

(D) obtuse angled

23. If $H_1, H_2, \dots, H_n$ are n harmonic means between a and b ($a \neq b$), then $\frac{H_1+a}{H_1-a} + \frac{H_n+b}{H_n-b} =$

(A) $2n$

(B) $n+1$

(C) $n-1$

(D) $2n+1$

24. If $y = \sin^{-1} \left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}} \right)$ ($x > 0$), then $\frac{dy}{dx} =$

(A) $\frac{x^2-1}{x^4+3x^2+1}$

(B) $\frac{x^2+1}{x^4+3x^2+1}$

(C) $\frac{x^2-1}{x^4-3x^2+1}$

(D) $\frac{x^2+1}{x^4-3x^2+1}$

25. If $32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha$ and $3 \cos 2\theta = 1$, then the general value of α for

(A) $n\pi \pm \frac{\pi}{3}$

(B) $2n\pi \pm \frac{2\pi}{3}$

(C) $2n\pi \pm \frac{\pi}{3}$

(D) $n\pi \pm \frac{2\pi}{3}$

26. In a ΔABC , $a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} = \frac{3b}{2}$, then the sides of the triangle are in

(A) AP

(B) GP

(C) HP

(D) None of the above

27. If θ is acute angle between the pair of lines $x^7 - 7xy + 12y^2 = 0$, then $\frac{2 \cos \theta + 3 \sin \theta}{4 \sin \theta + 5 \cos \theta} =$

(A) $\frac{29}{69}$

(B) $\frac{61}{39}$

(C) $\frac{39}{61}$

(D) $\frac{69}{29}$

28. The four geometric means between 2 and 64 are

(A) $\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}$

(B) 4, 8, 16, 32

(C) $4\sqrt{2}, 8, 16\sqrt{2}, 32$

(D) None of the above

29. The lines $px + qy = 1$ and $qx + py = 1$ are respectively the sides AB, AC of the ΔABC and the base BC is bisected at (p, q) . Equation of the median of the triangle through the vertex A is

(A) $(2pq - 1)(qx + py - 1) - (p^2 + q^2 - 1)(px +$





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

$$qy - 1) = 0$$

$$(B) (2pq - 1)(px + qy - 1) + (p^2 + q^2 - 1)(qx + py - 1) = 0$$

$$(C) (2pq - 1)(px + qy - 1) - (p^2 + q^2 - 1)(qx + py - 1) = 0$$

$$(D) (2pq - 1)(qx + py - 1) + (p^2 + q^2 - 1)(px + qy - 1) = 0$$

30. If the vectors $a\hat{i} + \hat{j} + \hat{k}$, $\hat{i} + b\hat{j} + \hat{k}$, $\hat{i} + \hat{j} + c\hat{k}$ ($a, b, c \neq 1$) are coplanar, then $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} =$

- (A) 0 (B) 1
(C) 2 (D) 3

31. Let $a = \hat{i} + \hat{j}$ and $b = 2\hat{i} - \hat{k}$. Then, the point of intersection of the lines $r \times a = b \times a$ and $r \times b = a \times b$ is

- (A) $-\hat{i} + \hat{j} + \hat{k}$ (B) $3\hat{i} - \hat{j} + \hat{k}$
(C) $\hat{i} - \hat{j} - \hat{k}$ (D) $3\hat{i} + \hat{j} - \hat{k}$

32. If the system of equations $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + \lambda z = -3$ has at least one solution, then λ is equal to

- (A) -5 (B) 3
(C) 5 (D) 6

33. If a variable takes value 0, 1, 2, ..., 50 with frequencies $1, {}^{50}C_1, {}^{50}C_2, \dots, {}^{50}C_{50}$, then the AM is

- (A) 50 (B) 25
(C) $\frac{2^{50}}{50}$ (D) 51

34. If $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$, then the number of elements in $(A \cup B) \times (A \cap B) \times (A \Delta B)$

- (A) 18 (B) 20
(C) 24 (D) 30

35. If n is an integer between 0 to 21, then find a value of n for which the value of $n! (21 - n)!$ is minimum.

- (A) 9 (B) 10
(C) 12 (D) 21

36. Suppose $A_1, A_2, A_3, \dots, A_{30}$ are 30 sets each with five elements and $B_1, B_2, \dots, B_n$ are n sets (each with three

elements) such that $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^n B_j = S$ and each element of S belongs to exactly ten of the A_i 's and exactly 9 of B_j 's. Then, n is equal to

- (A) 15 (B) 45
(C) 75 (D) 90

37. The eccentric angle of the extremities of latusrectum of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are given by

- (A) $\tan^{-1}\left(\pm \frac{ae}{b}\right)$ (B) $\tan^{-1}\left(\pm \frac{be}{e}\right)$
(C) $\tan^{-1}\left(\pm \frac{b}{ae}\right)$ (D) $\tan^{-1}\left(\pm \frac{a}{be}\right)$

38. If $\alpha \neq \beta$ and $\alpha^2 = 5\alpha - 3$, $\beta^2 = 5\beta - 3$, then the equation whose roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ is

- (A) $3x^2 - 25x + 3 = 0$ (B) $3x^2 + 5x + 3 = 0$
(C) $3x^2 - 5x + 3 = 0$ (D) $3x^2 - 19x + 3 = 0$

39. The probability that a man who is x years old will die in a year is p . Then, amongst n persons $A_1, A_2, \dots, A_n$ each x year old now, the probability that A_1 will die in one year is

- (A) $\frac{1}{n^2}$ (B) $1 - (1 - p)^n$
(C) $\frac{1}{n^2} [1 - (1 - p)^n]$ (D) $\frac{1}{n} [1 - (1 - p)^n]$

40. Angle between a and b is 120° . If $|b| = 2|a|$ and the vectors $a + xb$, $a - b$ are at right angle, then x is equal to

- (A) $\frac{1}{3}$ (B) $\frac{1}{5}$
(C) $\frac{2}{3}$ (D) $\frac{2}{5}$

41. If a number x is selected at random from natural numbers $1, 2, \dots, 100$, then the probability for $x + \frac{100}{x} > 29$ is

- (A) $\frac{37}{50}$ (B) $\frac{39}{50}$
(C) $\frac{41}{50}$ (D) $\frac{43}{50}$

42. If X and Y are two sets, then $X \cap Y' \cap (X \cup Y)'$ is

- (A) X' (B) Y'
(C) $\emptyset$ (D) None of these





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

43. $\int e^x (\sin hx + \cos hx) dx$ is equal to

- (A) $e^x \sec hx + C$ (B) $e^x \cos hx + C$
(C) $\sinh 2x + C$ (D) $\cosh 2x + C$

44. If $e_1 = (1, 1, 1)$ and $e_2 = (1, 1, -1)$ and a and b are two vectors such that $e_1 = 2a + b$ and $e_2 = a + 2b$, then angle between a and b is

- (A) $\cos^{-1} \left(-\frac{7}{11} \right)$ (B) $\cos^{-1} \left(\frac{7}{11} \right)$
(C) $\cos^{-1} \left(\frac{7}{9} \right)$ (D) $\cos^{-1} \left(\frac{6\sqrt{2}}{11} \right)$

45. If $P(1, 2)$, $Q(4, 6)$, $R(5, 7)$ and $S(a, b)$ are the vertices of a parallelogram $PQRS$, then

- (A) $a = 2, b = 3$ (B) $a = 3, b = 4$
(C) $a = 2, b = 4$ (D) $a = 3, b = 5$

46. If $a \cos \theta + b \sin \theta = 2$ and $a \sin \theta - b \cos \theta = 3$, then $a^2 + b^2$ is equal to

- (A) 6 (B) 5
(C) 13 (D) 10

47. The three thrown of three dice, the probability of throwing triplets not more than twice is

- (A) $1 - \frac{1}{6^2}$ (B) $1 - \frac{1}{6^3}$
(C) $1 - \frac{1}{36^2}$ (D) $1 - \frac{1}{36^3}$

48. $\int 3^{3^{3x}} \cdot 3^{3x} \cdot 3^x dx$ is equal to

- (A) $\frac{3^{3^{3x}} \cdot 3^x}{(\log 3)^3} + C$ (B) $\frac{3^x}{(\log 3)^3} + C$
(C) $\frac{3^{3^x}}{(\log 3)^3} + C$ (D) $\frac{3^{3^{3x}}}{(\log 3)^3} + C$

49. There are 50 questions in a paper. Find the number of ways in which a student can attempt one or more questions.

- (A) $2^{50} - 1$ (B) $2^{50} + 1$
(C) $2^{50} - 2$ (D) $2^{50} + 2$

50. Consider the following frequency distribution table.

Class interval	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Frequency	180	f_1	34	180	136	f_2	50

If the total frequency is 686 and the median is 42.6, then the value of f_1 and f_2 are

- (A) 81, 25 (B) 82, 24
(C) 83, 23 (D) 84, 22

Reasoning/Aptitude:

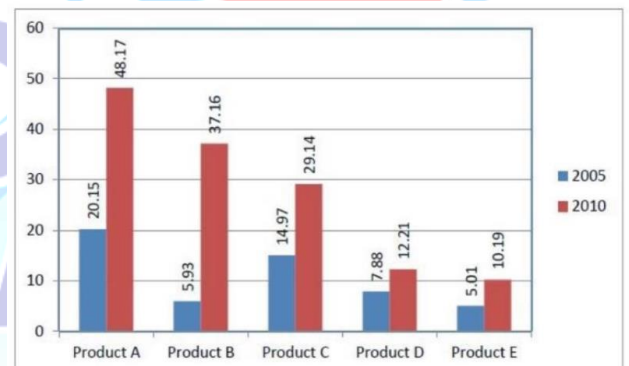
51. Choose the word which has the same relationship among the given three words:

Yellow, Blue, red

- (A) Black (B) Orange
(C) White (D) Paint

52. A company produces five different products. The sales of these five products (in lakh number of packs) during 2005 and 2010 are shown in the following bar-graph. Question is based on this graph.

Sales (in lakh numbers of packs) of five different products of a company during 2005-2010.



The sales of Product A in 2010 was by that percent more than the sales of Product B in 2010? (rounded off to the nearest integer)

- (A) 33% (B) 31%
(C) 28% (D) 22%





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

53. Choose the pair of number that comes next:

- 1 10 7 20 13 30 19
(A) 25 22 (B) 26 24
(C) 26 23 (D) 25 23

54. Read the following given below and answer the following questions:

- (i) In a Family of six persons A, B, C, D, E and F, there are two married couples.
(ii) D is grandmother of A and mother of B.
(iii) C is wife of B and mother of F.
(iv) F is the granddaughter of E.

Who among the following is one of the couples ?

- (A) CD (B) DE
(C) EB (D) None of these

55. If $Z = 52$ and $ACT = 48$, then BAT will be equal to

- (A) 46 (B) 39
(C) 44 (D) 41

Directions (Q. Nos. 56 to 57) Read the information given below and answer the following question.

- (i) In a family of six persons A, B, C, D, E and F there are two married couples.
(ii) D is grandmother of A and mother of B.
(iii) C is wife of B and mother of F.
(iv) F is the granddaughter of E.

61. Study the following table carefully and answer the following question.

Subjects (Full Marks 50)	40 and above	30 and above	20 and above	10 and above	0 and above
Physics	9	32	80	92	100
Chemistry	4	21	66	81	100
Aggregate	7	27	73	87	100

If it is known that atleast 23 students were eligible for a Symposium on Chemistry the minimum qualifying marks in Chemistry for eligibility to Symposium would lie in the range :

- (A) 30-40 (B) 20 - 30
(C) Below 20 (D) Cannot be determined

56. Which of the following is true?

- (A) A is brother of F. (B) A is sister of F.
(C) D has two grandsons. (D) None of these

57. What is C to A ?

- (A) Daughter (B) Grandmother
(C) Mother (D) Cannot be determined

58. Find the matching term for Music : Guitar : Performer.

- (A) Dance : Tune : Instrument
(B) Food : Recipe : Cook
(C) Patient : Medicine : doctor
(D) Trick : Rope : Acrobat

59. If $9 \times 3 + 8 = 24$, $10 \times 2 + 7 = 35$ and $80 \times 40 + 3 = 6$, then find the value of $12 \times 4 + 3 = ?$

- (A) 7 (B) 9
(C) 12 (D) 16

60. Choose the missing term out of the given alternatives.

Q1F, S2E, U6D, W21C, ?

- (A) Y66B (B) Y44B
(C) Y88B (D) Z88C





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

62. In a certain code language, "do re me" means "he is late"; "fa me la" means "she is early" and "so ti do" means "he leaves soon". Which word in the language means 'late'?

- (A) la (B) do
(C) me (D) re

63. Find the missing term $7 : 56 :: 9 : ?$.

- (A) 63 (B) 81
(C) 90 (D) 99

64. In this series, looking at the letter pattern fill the blank in the middle of the series :

ELFA, GLHA, ILJA,, MLNA

- (A) OLPA (B) KLMA
(C) LLMA (D) KLLA

65. Read the following information carefully and then answer the following question:

- (i) $P \Psi Q$ means P is mother of Q
(ii) $P \in Q$ means P is sister of Q
(iii) $P \$ Q$ means P is father of Q
(iv) $P \# Q$ means P is brother of Q

Which of the following means R is brother of T ?

- (A) $R \Psi S \# U \$ T$ (B) $U \Psi R \# S \# T$
(C) $U \Psi R \in S \Psi T$ (D) $T \# S \$ Q \in R$

66. Choose the pair that best represents a similar relationship to the one expressed in the original pair of words:

WAITRESS : RESTAURANT

- (A) doctor : diagnosis (B) actor : role
(C) driver : truck (D) teacher : school

67. COUNSEL is to BITIRAK, so also GUIDANCE is to

- (A) EOHYZKBB (B) FOHYZJBB
(C) FPHZZKAB (D) HOHYBJBA

68. Choose the number pair or group that is different from others ?

- (A) 15 : 46 (B) 12 : 37
(C) 9 : 28 (D) 8 : 33

69. If it is Saturday on January 1, 2000, then January 1, 2001 would have been

- (A) Monday (B) Sunday
(C) Tuesday (D) Friday

70. Radha is twice as old as Rita was 2 yr ago. If difference between their ages is 2 yr, how old is Radha today ?

- (A) 6 (B) 8
(C) 10 (D) 12

71. How many times in a day are the hands of a clock pointing opposite to each other?

- (A) 4 (B) 20
(C) 22 (D) 14

Direction (Q. 72 and 73) Study the following table carefully and answer the following questions.

Subjects (Full Marks 50)	40 and above	30 and above	20 and above	10 and above	0 and above
Physics	9	32	80	92	100
Chemistry	4	21	66	81	100
Agregate	7	27	73	87	100

72. If atleast 60% marks in physics are required for pursuing higher studies in physics. How many students will be eligible to pursue higher studies in physics ?

- (A) 27 (B) 32
(C) 34 (D) 41

73. The number of students scoring less than 40% marks in aggregate is

- (A) 13 (B) 19
(C) 27 (D) 20





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

74. Choose the missing term : BYCX : DWEV :: FUGT ?

- (A) EHIJ (B) GHIJ
(C) HSIR (D) SRHS

75. Read the following given below and answer the following questions:

- (i) In a Family of six persons A, B, C, D, E and F, there are two married couples.
(ii) D is grandmother of A and mother of B.
(iii) C is wife of B and mother of F.
(iv) F is the granddaughter of E.

How many male members are there in the family ?

- (A) Three (B) Two
(C) Cannot be determined (D) None of these

76. Read the following information carefully and then answer the following question:

- (i) $P \Psi Q$ means P is mother of Q
(ii) $P \in Q$ means P is sister of Q
(iii) $P \$ Q$ means P is father of Q
(iv) $P \# Q$ means P is brother of Q

If $K \Psi L \in M \# N$, then how K is related with N ?

- (A) Mother (B) Aunt
(C) Aunt (D) Grandmother

77. Choose the correct option for the remainder when $X = 1! + 2! + 3! + \dots + 100!$ is divided by 24.

- (A) 9 (B) 11
(C) 152 (D) 13

78. There are six members in a family. A is the father of D, E is the grandfather of D. B is the daughter-in-law of C. F is the uncle of D. What is the relationship of C with F ?

- (A) Sister (B) Mother-in-law
(C) Nephew (D) Data inadequate

79. Read the following information carefully and then answer the following question:

- (i) $P \Psi Q$ means P is mother of Q
(ii) $P \in Q$ means P is sister of Q
(iii) $P \$ Q$ means P is father of Q
(iv) $P \# Q$ means P is brother of Q

Which of the following means N is definitely daughter of K ?

- (A) $K \$ L \# M \# N$ (B) $M \Psi K \$ N \in L$
(C) $K \Psi M \# L \in N$ (D) $L \Psi K \$ N \# M$

80. Choose the word that is necessary part of the underlined word: **Harvest**

- (A) autumn (B) crop
(C) stockpile (D) tractor

81. Study the following table carefully and answer the following questions.

Subjects (Full Marks 50)	40 and above	30 and above	20 and above	10 and above	0 and above
Physics	9	32	80	92	100
Chemistry	4	21	66	81	100
Agregate	7	27	73	87	100

What is the difference between the students passed with 30 as cut – off marks in Chemistry and those passed with 30 as cut – off marks in aggregate?

- (A) 3 (B) 4 (C) 5 (D) 6



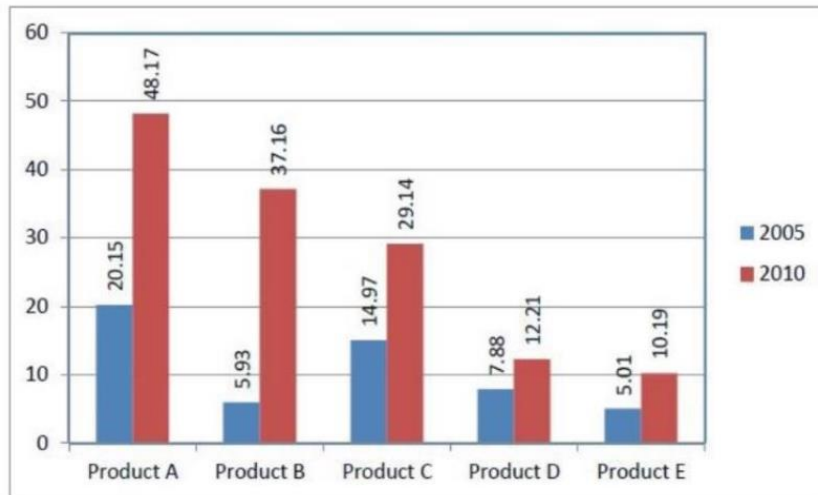


ACME ACADEMY

India's Most Trustable MCA Entrance Academy

82. A Company produces five different products. The sales of these five products (in lakh number of packs) during 2005 and 2010 are shown in the following bar – graph. Question is based on the graph.

Sales (in lakh number of packs) of five different products of a company during 2005-2010.



During the period 2005-2010. The minimum rate of increase in sales is in the case of

- (A) Product A
- (B) Product B
- (C) Product D
- (D) Product E

83. Insert the missing number 8, 7, 11, 12, 14, 17, 17, 22, ?

- (A) 27
- (B) 20
- (C) 22
- (D) 24

84. Choose the pair that best represents a similar relationship to the one expressed in the original pair of words:

PULSATE : THROB

- (A) walk : run
- (B) tired : sleep
- (C) examine : scrutinize
- (D) ballet : dancer

85. In a class of 50 students, Raghu's rank is twice that of Paul. There are 10 students who have ranked worse

than that of Raghu. Paul's rank in the class is

- (A) 5th
- (B) 10th
- (C) 15th
- (D) 20th

86. Choose the odd one:

- (A) Zebra
- (B) Rhinoceros
- (C) Hyena
- (D) Giraffe

87. If 'A3T15R' stands for 'ACTOR' and 'D1T5' stands for 'DATE'. How will you code 'ROTATE'?

- (A) R16T1T5
- (B) R15T1T5
- (C) R15T1T16
- (D) R15C1T7



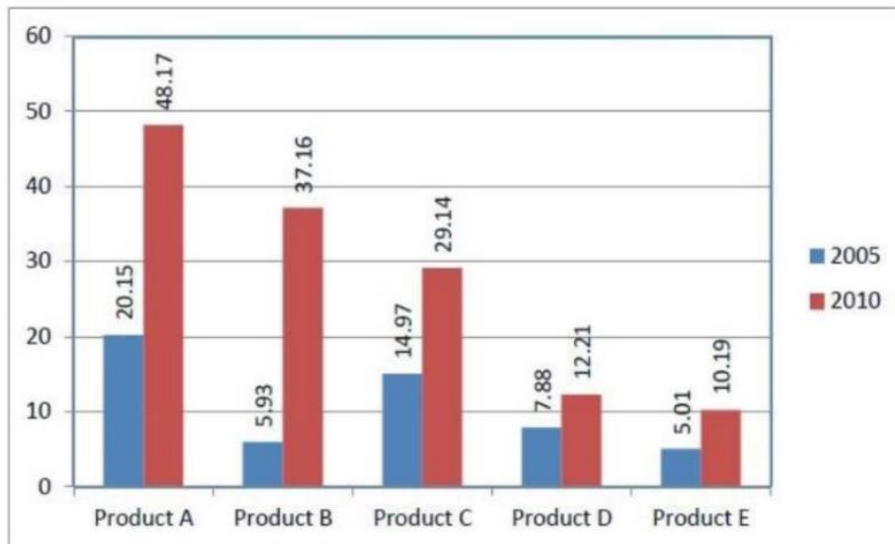


ACME ACADEMY

India's Most Trustable MCA Entrance Academy

88. A Company produces five different products. The sales of these five products (in lakh number of packs) during 2005 and 2010 are shown in the following bar – graph. Question is based on the graph.

Sales (in lakh number of packs) of five different products of a company during 2005-2010.



The sales have increased by nearly 55% from 2005 to 2010 in case of

- (A) Product A (B) Product B
(C) Product C (D) Product D

89. Which word does NOT belong with the others?

- (A) wing (B) fin
(C) beak (D) rudder

90. Insert the missing number: 16, 33, 65, 131, 261, ?

- (A) 523 (B) 521
(C) 613 (D) 721

Computer:

91. The process when processor fetch or decode another instruction during the execution of current instruction is called

- (A) Supercomputing (B) Pipelining
(C) Cloud Computing (D) Grid Computing

92. Which of the following is used by ALU to store the intermediate results?

- (A) Stack (B) Heap
(C) Registers (D) Accumulator

93. One TeraByte (TB) =GB and One ExaByte (EB) =GB

- (A) $2^{10}GB$, $2^{16}GB$ (B) $2^{10}GB$, $2^{20}GB$
(C) $2^{10}GB$, $2^{24}GB$ (D) $2^{10}GB$, $2^{30}GB$

94. The cache memory is more effective because of

- (A) Memory localisation
(B) Locality of reference
(C) Memory Size
(D) None of the mentioned





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

95. Which of the following is the fastest means of memory access for CPU?

- (A) Registers
- (B) Cache
- (C) Main Memory
- (D) Stack

96. The number $(2217)_8$ is equivalent to

- (A) $(608)_{16}$
- (B) $(028F)_{16}$
- (C) $(048F)_{16}$
- (D) $(2297)_{16}$

97. To fetch data from the secondary memory which one of the following registers is used?

- (A) MAR
- (B) PC
- (C) IR
- (D) MBR

98. The binary multiplication $00 * 11$ will give

- (A) 11
- (B) 00
- (C) 01
- (D) 10

99. Consider a computer system with speed of 10^6 instruction per second. A program P, having $2n^2$ steps is run on this system, where n is the input size. If $n = 10000$, what is the execution time for P?

- (A) 1.2 s
- (B) 20 s
- (C) 100 s
- (D) 200 s

100. To access the I/O devices, the status flag is continuously checking in

- (A) Program Controlled I/O
- (B) Memory mapped I/O
- (C) I/O mapped
- (D) None of the

English:

Direction (Q. Nos. 101-102) Read the following passage carefully and answer the question that follow:

It is said with truth that the function of a university is to prepare the young to take their place in human society. It must provide its members with the knowledge and skill necessary to make them efficient citizens. But is the whole duty of man exhausted by the acquisition of knowledge and professional training? Is a university only an institution for higher learning, a factory which churns out clerks and technicians able to run the machinery of the State? Mere knowledge which gratifies curiosity is different from culture which refines personality. Culture is not remembering a mass of serious details about the dates of birth of the great heroes of the world or the interesting names of the fastest ships which cross the Atlantic or entertaining odds and ends gathered from the latest who's who. A well known institution of this country has for its motto *sa vidya yavimuchyate*: that is, knowledge which is designed for salvation, for the development of the soul, is the best. Such an idea is not merely an Indian idiosyncrasy. Plato said long ago that the culture of soul is "the first and fairest thing that the

best of men can ever have. According to Goethe, the object of education is to form tastes and not simply to communicate knowledge. A man's culture is not judged by the amount of tabulated information which he has at his command, but by the quality of mind which he brings to bear on the facts of life. Education is not cramming the mind with a host of technical details, putting sight, as it were, into blind eyes. The eye of the soul is never blind, only its gaze may be turned to the false and the fleeting. Too often the vision may be dragged downwards by the "leaden weights" of pride and prejudice, of passion and desire. The function of the teacher is not to add the "leaden weights" but remove them and liberate the soul from the encumbrance so that it may follow its native impulse to soar upwards. The student at a university does not merely learn something, but becomes something by being exposed, in the most elastic period of his life, to transforming influences, such as the constant clash of mind with mind, the interchange of ideas, the testing of opinions, and the growth of knowledge of human nature.





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

101. Mere knowledge and culture may be distinguished from each other in that:

- (A) the former widens the mental horizon, the latter enlarges the hearts
- (B) the former gratifies curiosity, the latter refines personality
- (C) the former is concerned with facts, the latter with fiction
- (D) the former adds to power, the latter adds to prestige

102. What is the function of education according to the ancient Indian philosophers?

- (A) Education is the development
- (B) Education brings about salvation and development of the soul
- (C) Education is the cultivation of culture
- (D) Education aims at the inculcation of generosity

103. Choose the most suitable synonym of the given word:

SCINTILLATING

- (A) touching
- (B) nagging
- (C) glittering
- (D) warning

104. Choose the most suitable synonym of the given word:

BLAND

- (A) pleasant
- (B) harsh
- (C) irritating
- (D) tasteless

Directions (Q. Nos. 105-108) Read the following passage carefully and answer the question that follow:

It is said with truth that the function of a university is to prepare the young to take their place in human society. It must provide its members with the knowledge and skill necessary to make them efficient citizens. But is the whole duty of man exhausted by the acquisition of knowledge and professional training? Is a university only an institution for higher learning, a factory which churns out clerks and technicians able to run the machinery of the State? Mere knowledge which gratifies curiosity is different from culture which refines personality. Culture

is not remembering a mass of serious details about the dates of birth of the great heroes of the world or the interesting names of the fastest ships which cross the Atlantic or entertaining odds and ends gathered from the latest who's who. A well known institution of this country has for its motto *sa vidya yavimuchyate*: that is, knowledge which is designed for salvation, for the development of the soul, is the best. Such an idea is not merely an Indian idiosyncrasy. Plato said long ago that the culture of soul is "the first and fairest thing that the best of men can ever have. According to Goethe, the object of education is to form tastes and not simply to communicate knowledge. A man's culture is not judged by the amount of tabulated information which he has at his command, but by the quality of mind which he brings to bear on the facts of life. Education is not cramming the mind with a host of technical details, putting sight, as it were, into blind eyes. The eye of the soul is never blind, only its gaze may be turned to the false and the fleeting. Too often the vision may be dragged downwards by the "leaden weights" of pride and prejudice, of passion and desire. The function of the teacher is not to add the "leaden weights" but remove them and liberate the soul from the encumbrance so that it may follow its native impulse to soar upwards. The student at a university does not merely learn something, but becomes something by being exposed, in the most elastic period of his life, to transforming influences, such as the constant clash of mind with mind, the interchange of ideas, the testing of opinions, and the growth of knowledge of human nature.

105. What is meant by "leaden weights"?

- (A) Weights made of leaden
- (B) Cold scientific knowledge
- (C) Hindering factors, of pride and prejudice, passion, and desire, in the liberation of the soul
- (D) Social and family responsibilities

106. The function of university is

- (A) to enable the young to gather facts about the world
- (B) to learn to do his job





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

- (C) to prepare the young to take their place in society
(D) to enable them to learn to talk with others

107. According to the passage, the function of the teacher is

- (A) to remove leaden weights of pride and prejudice, passion, and desire to liberate the soul.
(B) to install facts into the minds of the students
(C) to teach humanity
(D) to foster brotherhood

108. What are the hindering factors in the liberation and development of the soul?

- (A) Power and wealth
(B) Passion and desire, pride and prejudice
(C) Money and influence
(D) Greed and envy

109. Fill in the blank with the most appropriate word given:

I cannot it to you right now; tomorrow we will discuss about it.

- (A) demand (B) disturb
(C) explain (D) recur

110. Choose the most suitable synonym of the given word?

RECUPERATE

- (A) recapture (B) disturb
(C) recover (D) expect

111. Fill in the blanks with the most appropriate word:

The _____ with which he is able to wield the paint – brush is really remarkable?

- (A) ease (B) practice
(C) sweep (D) gait

Directions (Q. Nos. 105-108) Read the following passage carefully and answer the question that follow:

It is said with truth that the function of a university is to prepare the young to take their place in human society. It must provide its members with the knowledge and skill necessary to make them efficient citizens. But is the

whole duty of man exhausted by the acquisition of knowledge and professional training? Is a university only an institution for higher learning, a factory which churns out clerks and technicians able to run the machinery of the State? Mere knowledge which gratifies curiosity is different from culture which refines personality. Culture is not remembering a mass of serious details about the dates of birth of the great heroes of the world or the interesting names of the fastest ships which cross the Atlantic or entertaining odds and ends gathered from the latest who's who. A well known institution of this country has for its motto *sa vidya yavimuchyate*: that is, knowledge which is designed for salvation, for the development of the soul, is the best. Such an idea is not merely an Indian idiosyncrasy. Plato said long ago that the culture of soul is "the first and fairest thing that the best of men can ever have. According to Goethe, the object of education is to form tastes and not simply to communicate knowledge. A man's culture is not judged by the amount of tabulated information which he has at his command, but by the quality of mind which he brings to bear on the facts of life. Education is not cramming the mind with a host of technical details, putting sight, as it were, into blind eyes. The eye of the soul is never blind, only its gaze may be turned to the false and the fleeting. Too often the vision may be dragged downwards by the "leaden weights" of pride and prejudice, of passion and desire. The function of the teacher is not to add the "leaden weights" but remove them and liberate the soul from the encumbrance so that it may follow its native impulse to soar upwards. The student at a university does not merely learn something, but becomes something by being exposed, in the most elastic period of his life, to transforming influences, such as the constant clash of mind with mind, the interchange of ideas, the testing of opinions, and the growth of knowledge of human nature.

112. What is the object of education according to Goethe?

- (A) It teaches social manners
(B) It teaches courtesy





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

- (C) It communicate knowledge
(D) It forms taste

113. What is the man's culture judged by?

- (A) By the quality of mind which he brings to bear on the facts of life
(B) By man's social skill
(C) By the variety of books he read
(D) By money and influence

114. Choose the most suitable antonym of given word:

ERUDITE

- (A) scholarly (B) unfamiliar
(C) illiterate (D) arrogant

115. Choose one word for the following

An uncivilized/primitive person.

- (A) barbarian (B) ascetic
(C) bourgeois (D) altruist

116. Fill in the blank with the most appropriate word.

The car driver was arrested for rash driving and his license was by the police.

- (A) impounded
(C) penalised

- (B) flown
(D) banned

117. Fill in the blank with the most appropriate word:
The State Transportation Corporation has a loss of 5 crore INR this year.

- (A) derived (B) incurred
(C) performed (D) formulated

118. Choose the most suitable antonym of given word.

NEGLIGENCE

- (A) diligence (B) meticulousness
(C) integrity (D) honest

119. Write one word for the following.

One who does not save for future.

- (A) incorporeal (B) inedible
(C) improvident (D) implacable

120. Fill in the blank with the most appropriate word given in the options.

The child prefers rooms with ceilings.

- (A) tall (B) long
(C) higher (D) high





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

Answer Key

1. C	13. C	25. B	37. C	49. A	61. B	73. C	85. D	97. A	109. C
2. C	14. D	26. A	38. D	50. B	62. D	74. C	86. C	98. B	110. C
3. B	15. C	27. A	39. D	51. B	63. C	75. C	87. B	99. D	111. C
4. A	16. C	28. B	40. D	52. B	64. D	76. A	88. D	100. A	112. D
5. B	17. C	29. C	41. B	53. *	65. B	77. A	89. C	101. B	113. A
6. B	18. C	30. B	42. C	54. B	66. D	78. D	90. A	102. B	114. C
7. B	19. C	31. D	43. B	55. A	67. B	79. B	91. B	103. C	115. A
8. B	20. B	32. A	44. A	56. D	68. D	80. B	92. D	104. D	116. A
9. C	21. C	33. B	45. A	57. C	69. A	81. D	93. D	105. C	117. B
10. D	22. D	34. D	46. C	58. D	70. B	82. C	94. B	106. C	118. B
11. A	23. A	35. B	47. D	59. B	71. C	83. B	95. A	107. A	119. C
12. B	24. A	36. B	48. D	60. C	72. B	84. C	96. C	108. B	120. D

ACME
MCA
ENTRANCE ACADEMY





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

Solution

1. (c) Given, $|k| = 5$ where $k = 3\cos\theta + 4\sin\theta$

$$\Rightarrow |3\cos\theta + 4\sin\theta| = 5 \Rightarrow \left| 5\left(\frac{3}{5}\cos\theta + \frac{4}{5}\sin\theta\right) \right| = 5$$

Let $\cos\alpha = \frac{3}{5} \Rightarrow \sin\alpha = \frac{4}{5}$

$$\therefore |5(\cos\theta \cdot \cos\alpha + \sin\theta \cdot \sin\alpha)| = 5$$

$$\Rightarrow |5(\cos(\theta - \alpha))| = 5 \Rightarrow 5|\cos(\theta - \alpha)| = 5$$

$$\Rightarrow \cos(\theta - \alpha) = \pm 1 \Rightarrow \theta - \alpha = 0^\circ, 180^\circ$$

$$\Rightarrow \theta = \alpha, \theta = \alpha + 180^\circ$$

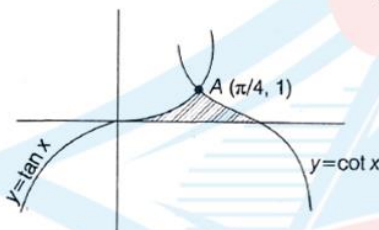
$\Rightarrow$ It has two solutions in it.

2. (c) Given curves are $\tan x, -\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$

and $y = \cot x, \frac{\pi}{6} \leq x \leq \frac{3\pi}{2}$

On solving Eqs. (i) and (ii),

$$\tan x = \cot x \Rightarrow \tan^2 x = 1 = \tan^2\left(\frac{\pi}{4}\right)$$



$$\Rightarrow x = \frac{\pi}{4}$$

$$\begin{aligned} \therefore \text{Required area} &= \int_0^{\pi/4} \tan x dx + \int_{\pi/4}^{\pi/2} \cot x dx \\ &= [(\log \sec x)]_0^{\pi/4} + [(\log \sin x)]_{\pi/4}^{\pi/2} \\ &= (\log \sqrt{2} - \log 1) + \left(\log 1 - \log\left(\frac{1}{\sqrt{2}}\right) \right) \\ &= 2\log \sqrt{2} = \log 2 \end{aligned}$$

3. (b) If $\log(1 - x + x^2) = a_1x + a_2x^2 + a_3x^3 + \dots$... (i)

To find $a_3 + a_6 + a_9 + \dots$

In Eq. (i), put $x = 1$

$$\log(1) = a_1 + a_2 + a_3 + a_4 + \dots \quad \dots (ii)$$

In Eq. (i), put $x = \omega$

$$\log(1 - \omega + \omega^2) = a_1\omega + a_2\omega^2 + a_3\omega^3 + \dots \quad \dots (iii)$$

In Eq. (i), put $x = \omega^2$

$$\log(1 + \omega - \omega^2) = a_1\omega^2 + a_2\omega + a_3 + \dots \quad \dots (iv)$$

$$\text{where, } \omega = \frac{-1 + \sqrt{3}i}{2} \text{ and } \omega^2 = \frac{-1 - \sqrt{3}i}{2}$$

$$[\because 1 + \omega + \omega^2 = 0 \text{ and } \omega^3 = 1]$$

On adding Eqs. (ii), (iii) and (iv),

$$\log 1 + \log(1 - \omega + \omega^2) + \log(1 + \omega - \omega^2)$$

$$= a_1(1 + \omega + \omega^2) + a_2(1 + \omega + \omega^2) + 3a_3 + 3a_6 + \dots$$

$$\Rightarrow 0 + \log(-2\omega) + \log(-2\omega^2) = a_1(0) + a_2(0) + 3(a_3 + a_6 + a_9 + \dots)$$

$$= \frac{\log(4\omega^3)}{3} = a_3 + a_6 + a_9 + \dots = a_3 + a_6 + a_9 + \dots = \frac{2}{3} \log 2$$

4. (a) We have, $\lim_{x \rightarrow \infty} \left(\frac{x+7}{x+2} \right)^{x+5} = \lim_{x \rightarrow \infty} \left(1 + \frac{5}{x+2} \right)^{x+5}$ [1[∞] form]

$$= \lim_{x \rightarrow \infty} \left[\frac{5}{x+2} \right]^{x+5} = e^{\lim_{x \rightarrow \infty} \frac{5x(1 + \frac{5}{x})}{x(1 + \frac{2}{x})}} = e^{\lim_{x \rightarrow \infty} 5 \left(\frac{1 + \frac{5}{x}}{1 + \frac{2}{x}} \right)} = e^{5 \left(\frac{1+0}{1+0} \right)} = e^5$$

5. (b) Given circles are $x^2 + y^2 = 4$... (i)

Centre $C_1 = (0, 0)$ and radius $(r_1) = 2$

and $x^2 + y^2 - 6x - 8y = 24$

$$\Rightarrow x^2 - 6x + 9 + y^2 - 8y + 16 = 24 + 9 + 16$$

$$\Rightarrow (x-3)^2 + (y-4)^2 = (7)^2$$

$\therefore$ Center $(C_2) = (3, 4)$ and radius $(r_2) = 7$

$$\therefore C_1C_2 = \sqrt{(3-0)^2 + (4-0)^2} = 5$$

and $r_1 + r_2 = 2 + 7 = 9$

$$\therefore C_1C_2 < r_1 + r_2 \text{ but } C_1C_2 = |r_1 - r_2|$$

Thus, the given circles touch each other internally.

Hence, number of common tangent is only one.

6. (b) Given that $\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} = k$

$$\text{Let } \tan \frac{A}{2} = x, \tan \frac{B}{2} = y, \tan \frac{C}{2} = z$$

$$\therefore x^2 + y^2 + z^2 - xy - yz - zx$$

$$= \frac{1}{2}[(x-y)^2 + (y-z)^2 + (z-x)^2] \geq 0$$

$$\Rightarrow x^2 + y^2 + z^2 - xy - yz - zx \geq 0$$

$$\Rightarrow (x^2 + y^2 + z^2) \geq xy + yz + zx$$

$$\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq \tan \frac{A}{2} \cdot \tan \frac{B}{2} + \tan \frac{B}{2} \cdot \tan \frac{C}{2} + \tan \frac{C}{2} \cdot \tan \frac{A}{2} \quad \dots (i)$$

$$\therefore A + B + C = 180^\circ$$

$$\therefore \frac{A}{2} + \frac{B}{2} + \frac{C}{2} = 90^\circ$$

$$\frac{A}{2} + \frac{B}{2} = 90^\circ - \frac{C}{2} \Rightarrow \tan\left(\frac{A}{2} + \frac{B}{2}\right) = \tan\left(90^\circ - \frac{C}{2}\right)$$

$$\Rightarrow \frac{\tan \frac{A}{2} + \tan \frac{B}{2}}{1 - \tan \frac{A}{2} \cdot \tan \frac{B}{2}} = \cot \frac{C}{2}$$

$$\Rightarrow \tan \frac{A}{2} \cdot \tan \frac{C}{2} + \tan \frac{B}{2} \cdot \tan \frac{C}{2} = 1 - \tan \frac{A}{2} \cdot \tan \frac{B}{2}$$

$$\Rightarrow \tan \frac{A}{2} \cdot \tan \frac{B}{2} + \tan \frac{B}{2} \cdot \tan \frac{C}{2} + \tan \frac{C}{2} \cdot \tan \frac{A}{2} = 1$$

From Eq. (i),

$$\tan^2 \frac{A}{2} + \tan^2 \frac{B}{2} + \tan^2 \frac{C}{2} \geq 1 \text{ or } k \geq 1$$

7. (b) Given

$$\mathbf{a} = 2\hat{i} + \hat{j} + 2\hat{k}, \mathbf{b} = \hat{i} - \hat{j} + 2\hat{k} \text{ and } \mathbf{c} = \hat{i} + \hat{j} - 2\hat{k}$$

Let $\mathbf{d}$ be the vector in the plane of $\mathbf{a}$ and $\mathbf{c}$





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

$$\therefore \mathbf{d} = \mathbf{a} + 6\mathbf{c} = (2\hat{i} + \hat{j} + 2\hat{k}) + \lambda(\hat{i} + \hat{j} - 2\hat{k})$$

$$\mathbf{d} = (2 + \lambda)\hat{i} + (1 + \lambda)\hat{j} + (2 - 2\lambda)\hat{k}$$

Now, projection of $\mathbf{d}$ on $\mathbf{b} = \frac{\mathbf{d} \cdot \mathbf{b}}{|\mathbf{b}|}$

$$\frac{1}{\sqrt{6}} = \frac{(2 + \lambda) - (1 + \lambda) + 2(2 - 2\lambda)}{\sqrt{1^2 + (-1)^2 + 2^2}} \Rightarrow \frac{1}{\sqrt{6}} = \frac{5 - 4\lambda}{\sqrt{6}}$$

$$\therefore 5 - 4\lambda = 1 \Rightarrow \lambda = 1$$

$$\therefore \mathbf{d} = (2 + 1)\hat{i} + (1 + 1)\hat{j} + (2 - 2)\hat{k} \Rightarrow \mathbf{d} = 3\hat{i} + 2\hat{j}$$

8. (b) Given polynomial

$$p(x) = x^4 - 3x^3 + 2px^2 - 6$$

$\therefore p(x)$ is divisible by $(x - 1)$

$$\Rightarrow p(1) = 0 \Rightarrow 1^4 - 3(1)^3 + 2p(1)^2 - 6 = 0$$

$$\Rightarrow 8 = 2p \Rightarrow p = 4$$

$$9. (c) F(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \therefore F(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore F(\theta) \cdot F(\alpha) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta \cdot \cos\alpha - \sin\theta \sin\alpha & -(\cos\theta \sin\alpha + \sin\theta \cos\alpha) & 0 \\ \sin\theta \cdot \cos\alpha + \cos\theta \sin\alpha & \cos\theta \cos\alpha - \sin\theta \sin\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos(\theta + \alpha) & -\sin(\theta + \alpha) & 0 \\ \sin(\theta + \alpha) & \cos(\theta + \alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

[Using $\cos(A + B) = \cos A \cos B - \sin A \sin B$
 $\sin(A + B) = \sin A \cos B + \cos A \sin B$]

$$= F(\theta + \alpha)$$

10. (d) Given that

$$\frac{n!}{2!(n-2)!} : \frac{n!}{4!(n-4)!} = 2 : 1$$

$$\Rightarrow \frac{2!(n-2)(n-3)(n-4)!}{n!} = \frac{2}{1} \Rightarrow \frac{12}{n^2 - 5n + 6} = 2$$

$$\Rightarrow n^2 - 5n = 0 \Rightarrow n = 5, n \neq 0$$

11. (a) Given that,

$$\text{Ellipse } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad \dots(i)$$

Equation of tangents on ellipse (i)

$$y = mx \pm \sqrt{a^2m^2 + b^2}$$

$$\Rightarrow (y - mx)^2 = a^2m^2 + b^2$$

$$\Rightarrow y^2 + m^2x^2 - 2mxy = a^2m^2 + b^2$$

$$\Rightarrow m^2(x^2 - a^2) - 2mxy + y^2 - b^2 = 0$$

which are perpendicular.

$$\therefore m_1 \cdot m_2 = 1 \Rightarrow \frac{y^2 - b^2}{x^2 - a^2} = -1$$

$$\Rightarrow x^2 + y^2 = a^2 + b^2 \text{ which is an equation of circle.}$$

12. (b) Given that

$$\mathbf{OA} = \hat{i} - 4\hat{j} + 3\hat{k} \text{ and } \mathbf{OB} = -\hat{i} + 2\hat{j} - \hat{k}$$

Since, the median will pass through the mid-point of AB .

$$\text{Hence, median} = \frac{\mathbf{OA} + \mathbf{OB}}{2}$$

$$= \frac{(\hat{i} - 4\hat{j} + 3\hat{k}) + (-\hat{i} + 2\hat{j} - \hat{k})}{2} = -\hat{j} + \hat{k}$$

$$13. (c) \sin\theta = \frac{-1}{2}, \tan\theta = \frac{1}{\sqrt{3}}$$

Since, $\sin\theta$ is negative and $\tan\theta$ is positive.

$\Rightarrow \theta$ lies in third quadrant.

$$\therefore \sin\theta = \frac{-1}{2} = \sin\left(\pi + \frac{\pi}{6}\right) = \sin\frac{7\pi}{6}$$

$$\text{and } \tan\theta = \frac{1}{\sqrt{3}} = \tan\left(\pi + \frac{\pi}{6}\right) = \tan\frac{7\pi}{6}$$

$$\theta = 2n\pi \pm \left(\frac{7\pi}{6}\right), n \in I$$

14. (d) Let $\mathbf{OA} = 3\hat{i} + \hat{j}$

$$\mathbf{OB} = 5\hat{i} + 2\hat{j} + \hat{k} \Rightarrow \mathbf{OC} = \hat{i} - 2\hat{j} + 3\hat{k}$$

$$\therefore \mathbf{AB} = \mathbf{OB} - \mathbf{OA} = 2\hat{i} + \hat{j} + \hat{k}$$

$$\text{and } \mathbf{AC} = \mathbf{OC} - \mathbf{OA} = -\hat{i} - 3\hat{j} + 3\hat{k}$$

$$\therefore \text{Area of triangle} = \frac{1}{2} |\mathbf{AB} \times \mathbf{AC}| \quad \dots(i)$$

$$\mathbf{AB} \times \mathbf{AC} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ -1 & -3 & 3 \end{vmatrix} = \hat{i}(6) - \hat{j}(8) + \hat{k}(-4) = 6\hat{i} - 8\hat{j} - 4\hat{k}$$

$$\therefore \text{Area of triangle} = \frac{1}{2} \times \sqrt{6^2 + (-8)^2 + (-4)^2} = \sqrt{29} \text{ sq. units}$$

15. (c) Let the numbers are x_i , where $1 \leq i \leq 20$

$$\therefore \text{Mean } (\bar{x}) = \frac{\sum_{i=1}^{20} x_i}{20} \text{ Then, } \sigma(x) = \sqrt{\frac{\sum_{i=1}^{20} (x_i - \bar{x})^2}{n}} = 30 \quad [\text{given}]$$

Let the new numbers are y_i , where $y_i = x_i + 4$

$$\text{Then, } \bar{y} = \frac{\sum_{i=1}^{20} (x_i + 4)}{20} \Rightarrow \bar{y} = \bar{x} + 4$$

$$\therefore y_i - \bar{y} = (x_i + 4) - (\bar{x} + 4)$$

$$y_i - \bar{y} = x_i - \bar{x}$$

$$\therefore \sigma(y) = \sqrt{\frac{\sum_{i=1}^{20} (y_i - \bar{y})^2}{n}} = \sqrt{\frac{\sum_{i=1}^{20} (x_i - \bar{x})^2}{n}} = 30$$

$$16. (c) f(x) = \frac{x}{1 + x \tan x}, 0 \leq x \leq \frac{\pi}{2}$$

$$\therefore f'(x) = \frac{(1 + x \tan x) \cdot 1 - x(x \sec^2 x + \tan x)}{(1 + x \tan x)^2} = \frac{1 - x^2 \sec^2 x}{(1 + x \tan x)^2}$$

For maxima / minima

$$f'(x) = 0 \Rightarrow 1 - x^2 \sec^2 x = 0 \Rightarrow x^2 = \cos^2 x \Rightarrow x = \cos x$$

$\therefore f''(x) < 0$ at $x = \cos x$

$\Rightarrow f(x)$ is maximum when $x = \cos x$





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

17. (c) $f: R \rightarrow R$

$$f(x) = \begin{cases} \frac{x+2}{x^2+3x+2}, & \text{if } x \in R - \{-1, -2\} \\ -1, & \text{if } x = -2 \\ 0, & \text{if } x = -1 \end{cases}$$

$\therefore f(x)$ is continuous for every value of R except $\{-1, -2\}$.

Now, we have to check continuity at $x = -2$ and $x = -1$

$$\begin{aligned} \therefore \lim_{x \rightarrow -2} f(x) &= \lim_{x \rightarrow -2} \frac{x+2}{x^2+3x+2} \\ &= \lim_{x \rightarrow -2} \frac{(x+2)}{(x+1)(x+2)} = \frac{1}{-2+1} = -1 = f(-2) \end{aligned}$$

$\Rightarrow f(x)$ is continuous at $x = -2$

Now at $x = -1$

$$\lim_{x \rightarrow -1} f(x) = \lim_{x \rightarrow -1} \frac{(x+2)}{(x+1)(x+2)} = \lim_{x \rightarrow -1} \frac{1}{x+1} = \infty$$

Limit does not exist.

$\Rightarrow f(x)$ is not continuous at $x = -1$

18. (c) $\therefore$ Number of diagonals in n -sided polygon is $({}^nC_2 - n)$.

$$\therefore {}^nC_2 - n = 44 \Rightarrow \frac{n(n-1)}{2} - n = 44$$

$$\Rightarrow n(n-3) = 88 \Rightarrow n^2 - 3n - 88 = 0$$

$$\Rightarrow (n-11)(n+8) = 0 \Rightarrow n = 11, n \neq -8 \text{ (never be negative)}$$

19. (c) Given that

$$P(E) = 0.25, P(F) = 0.50 \Rightarrow P(E \cap F) = 0.14$$

To find $P(E \cup F)$

$$\therefore P(E \cup F) = P(E) + P(F) - P(E \cap F)$$

$$= 0.25 + 0.50 - 0.14 = 0.61$$

$$\therefore P(E \cup F)' = 1 - P(E \cup F) = 1 - 0.61 = 0.39$$

20. (b) $y = \tan^{-1} \left(\frac{3x - x^3}{1 - 3x^2} \right)$

Let $x = \tan \theta$

$$\therefore y = \tan^{-1} \left(\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \right) = \tan^{-1} (\tan 3\theta) = 3\theta$$

$$\Rightarrow y = 3 \tan^{-1} x \quad [\theta = \tan^{-1} x]$$

$$\therefore \frac{dy}{dx} = 3 \frac{d}{dx} (\tan^{-1} x) = \frac{3}{1+x^2}$$

21. (c) $\tan 9^\circ - \tan 27^\circ - \tan 63^\circ + \tan 81^\circ$

$$\begin{aligned} &= (\tan 9^\circ + \tan 81^\circ) - (\tan 27^\circ + \tan 63^\circ) \\ &= \frac{2}{\sin 18^\circ} - \frac{2}{\sin 54^\circ} = \frac{2}{(\sqrt{5}-1)} - \frac{2}{(\sqrt{5}+1)} = \frac{16}{5-1} = 4 \end{aligned}$$

22. (d) Let sides of triangles be a, b and c .

$$\therefore b + c = x, bc = y, a = z \text{ (Given)}$$

$$\text{and } y = (x-z)(x+z)$$

$$\Rightarrow bc = (b+c+a)(b+c-a)$$

$$\Rightarrow b^2 + c^2 - a^2 = -bc \Rightarrow \frac{b^2 + c^2 - a^2}{2bc} = \frac{-1}{2}$$

$$\Rightarrow \cos A = \frac{-1}{2} = \cos 120^\circ \Rightarrow A = 120^\circ$$

Hence, the triangle will be obtuse angled triangle.

23. (a) Given $H_1, H_2, \dots, H_n$ are Harmonic means between a and b , ($a \neq b$).

$$\Rightarrow \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \dots, \frac{1}{H_n} \text{ are Arithmetic mean between } \frac{1}{a} \text{ and } \frac{1}{b}.$$

$$\therefore \text{Common difference } (d) = \frac{1}{H_1} - \frac{1}{a} = \frac{1}{b} - \frac{1}{H_n}$$

$$\Rightarrow \frac{1}{a} = \frac{dH_1}{a-H_1} \Rightarrow a = \frac{a-H_1}{dH_1}$$

$$\text{Also, } b = \frac{b-H_n}{dH_n}$$

$$\begin{aligned} \text{Now, } \frac{H_1+a}{H_1-a} + \frac{H_n+b}{H_n-b} &= \frac{1}{d} \left(\frac{1}{b} - \frac{1}{a} + \frac{1}{H_n} - \frac{1}{H_1} \right) \\ &= (n+1) + (n-1) \quad \left[\because \frac{1}{b} - \frac{1}{a} = d(n+1) \right] \\ &= 2n \end{aligned}$$

24. (a) $y = \sin^{-1} \left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}} \right)$

$$\therefore \frac{dy}{dx} = \frac{1}{\sqrt{1-\frac{x^4+1+2x^2}{1+3x^2+x^4}}} \cdot \frac{d}{dx} \left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}} \right) = \frac{\sqrt{1+3x^2+x^4}}{x}$$

$$\begin{aligned} &\left\{ \frac{\sqrt{1+3x^2+x^4}(2x) - (x^2+1) \frac{1}{2\sqrt{1+3x^2+x^4}}(6x+4x^3)}{(1+3x^2+x^4)} \right\} \\ &= \frac{\sqrt{1+3x^2+x^4}}{x(1+3x^2+x^4)} \left\{ \frac{2x(1+3x^2+x^4) - (x^2+1)(3x+2x^3)}{\sqrt{1+3x^2+x^4}} \right\} \\ &= \frac{1}{x(1+3x^2+x^4)} \{ 2x + 6x^3 + 2x^5 - 3x^3 - 2x^5 - 3x - 2x^3 \} \\ &= \frac{1}{x(1+3x^2+x^4)} \{ 4x^3 - x \} \end{aligned}$$

$$\frac{dy}{dx} = \frac{x^2-1}{1+3x^2+x^4}$$

25. (b) Given that

$$32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha, 3 \cos 2\theta = 1$$

$$\therefore 3 \cos 2\theta = 1 \Rightarrow \cos 2\theta = 1/3$$

$$\Rightarrow 2 \cos^2 \theta - 1 = \frac{1}{3} \Rightarrow 2 \cos^2 \theta = \frac{2}{3} \Rightarrow \cos \theta = \sqrt{\frac{2}{3}}$$

$$\therefore \tan^2 \theta = \sec^2 \theta - 1 \Rightarrow \frac{3}{2} - 1 = \frac{1}{2}$$

$$\text{Now, } 32 \tan^8 \theta = 2 \cos^2 \alpha - 3 \cos \alpha$$

$$32 \left(\frac{1}{2} \right)^4 = 2 \cos^2 \alpha - 3 \cos \alpha$$

$$\Rightarrow 2 \cos^2 \alpha - 3 \cos \alpha - 2 = 0$$

$$\Rightarrow 2 \cos^2 \alpha - 4 \cos \alpha + \cos \alpha - 2 = 0$$

$$\Rightarrow 2 \cos \alpha (\cos \alpha - 2) + 1(\cos \alpha - 2) = 0$$

$$\Rightarrow (\cos \alpha - 2)(2 \cos \alpha + 1) = 0$$

$$\therefore \cos \alpha = \frac{-1}{2} \text{ or } \cos \alpha = 2 \text{ which is not possible.}$$

$$\cos \alpha = \cos \frac{2\pi}{3} \therefore \alpha = 2n\pi \pm \frac{2\pi}{3}$$





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

26. (a) Given that

$$\begin{aligned} a \cos^2 \frac{C}{2} + c \cos^2 \frac{A}{2} &= \frac{3b}{2} \\ \Rightarrow 2a \cos^2 \frac{C}{2} + 2c \cos^2 \frac{A}{2} &= 3b \\ \Rightarrow a(1 + \cos C) + c(1 + \cos A) &= 3b \\ \Rightarrow a + c + (a \cos C + c \cos A) &= 3b \\ \Rightarrow a + c + b &= 3b \Rightarrow a + c = 2b \Rightarrow a, b, c \text{ are in A.P.} \end{aligned}$$

27. (a) $x^2 - 7xy + 12y^2 = 0$

On comparing with $ax^2 + 2hxy + by^2 = 0$

$$h = \frac{-7}{2}, a = 1, b = 12$$

$$\therefore \tan \theta = \frac{2\sqrt{h^2 - ab}}{a + b} \Rightarrow \tan \theta = \frac{2\sqrt{\frac{49}{4} - 12}}{1 + 12} = \frac{1}{13}$$

$$\therefore \frac{2\cos\theta + 3\sin\theta}{4\sin\theta + 5\cos\theta} = \frac{2 + 3\tan\theta}{4\tan\theta + 5} = \frac{2 + 3\left(\frac{1}{13}\right)}{4\left(\frac{1}{13}\right) + 5} = \frac{29}{69}$$

28. (b) Let four geometric mean between 2 and 64 are G_1, G_2, G_3, G_4 .

$\therefore 2, G_1, G_2, G_3, G_4, 64$ from GP.

First term (a) = 2 and sixth term = 64

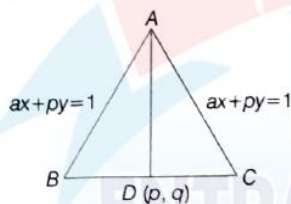
$ar^5 = 64$, where r is common ratio

$$2 \cdot r^5 = 64 \Rightarrow r^5 = 32 \Rightarrow r = 2$$

$$\begin{aligned} \therefore G_1 &= ar = 2 \times 2 = 4 \\ G_2 &= ar^2 = 2(2)^2 = 8 \\ G_3 &= ar^3 = 2(2)^3 = 16 \\ G_4 &= ar^4 = 2(2)^4 = 32 \end{aligned}$$

29. (c)

$$\begin{aligned} px + qy &= 1 & \dots(i) \\ qx + py &= 1 & \dots(ii) \end{aligned}$$



On solving Eqs. (i) and (ii),

$$px + qy = qx + py \Rightarrow x(p - q) + y(q - p) = 0 \Rightarrow x = y$$

From Eq. (i);

$$px + qx = 1 \Rightarrow x = \frac{1}{p + q} = y$$

$$\therefore A = \left(\frac{1}{p + q}, \frac{1}{p + q} \right)$$

$\therefore$ Equation of AD,

$$y - q = \frac{\frac{1}{p + q} - q}{\frac{1}{p + q} - p} (x - p)$$

$$\Rightarrow (y - q) = \left(\frac{1 - p^2 - pq}{p + q} \right) = \left(\frac{1 - pq - q^2}{p + q} \right) (x - p)$$

$$\begin{aligned} \Rightarrow y - yp^2 - pqy - q + qp^2 + pq^2 \\ = x - xq^2 - xpq - p + p^2q + pq^2 \\ \Rightarrow pqx + q^2x - p^2y - pqy + y - x + p - q = 0 \\ \Rightarrow (p + q)qx - py(p + q) + y - x + (p - q) = 0 \end{aligned}$$

On multiplying $py(p - q)$, we get

$$\begin{aligned} (p^2 - q^2)qx - (p^2 - q^2)py + y(p - q) - x(p - q) + (p - q)^2 &= 0 \\ \Rightarrow (2pq - 1)(px + qy - 1) &= (p^2 + q^2 - 1)(qx + py - 1) \\ \Rightarrow (2pq - 1)(px + qy - 1) - (p^2 + q^2 - 1)(qx + py - 1) &= 0 \end{aligned}$$

30. (b) Since $(a\hat{i} + \hat{j} + \hat{k})$, $(\hat{i} + b\hat{j} + \hat{k})$ and $(\hat{i} + \hat{j} + c\hat{k})$ are coplanar.

$$\therefore \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

Applying $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$,

$$\begin{vmatrix} a & 1 & 1 \\ 1 - a & b - 1 & 0 \\ 1 - a & 0 & c - 1 \end{vmatrix} = 0$$

$$\Rightarrow (1 - a)(1 - b)(1 - c) \begin{vmatrix} a & 1 & 1 \\ 1 - a & 1 - b & 1 - c \\ 1 & 0 & -1 \end{vmatrix} = 0$$

$$\Rightarrow \frac{a}{1 - a} + \frac{1}{1 - a} + \frac{1}{1 - c} = 0 \Rightarrow \frac{a}{1 - a} + 1 + \frac{1}{1 - b} + \frac{1}{1 - c} = 1$$

$$\Rightarrow \frac{1}{1 - a} + \frac{1}{1 - b} + \frac{1}{1 - c} = 1$$

31. (d) $\mathbf{a} = \hat{i} + \hat{j}$, $\mathbf{b} = 2\hat{i} - \hat{k}$

$$\mathbf{r} \times \mathbf{a} = \mathbf{b} \times \mathbf{a} \Rightarrow (\mathbf{r} - \mathbf{b}) \times \mathbf{a} = 0 \Rightarrow \mathbf{r} = \mathbf{b} + t\mathbf{a} \quad \dots(i)$$

$$\text{and } \mathbf{r} \times \mathbf{b} = \mathbf{a} \times \mathbf{b} \Rightarrow (\mathbf{r} - \mathbf{a}) \times \mathbf{b} = 0 \Rightarrow \mathbf{r} = \mathbf{a} + s\mathbf{b} \quad \dots(ii)$$

where s, t are scalars.

Eqs. (i) and (ii) are same if $s = t = 1$

$\therefore$ Point of intersection is $\mathbf{r} = \mathbf{a} + \mathbf{b}$

$$\Rightarrow \mathbf{r} = (\hat{i} + \hat{j}) + (2\hat{i} - \hat{k}) = 3\hat{i} + \hat{j} - \hat{k}$$

32. (a) Given system of equations

$$\begin{aligned} 3x - y + 4z &= 3 \\ x + 2y - 3z &= -2 \\ 6x + 5y + \lambda z &= -3 \end{aligned}$$

$$\Delta = \begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix}$$

$$\Delta = 30 - 65 - 7\lambda = -35 - 7\lambda$$

$\therefore$ System of equations has infinite solution if $\Delta = 0$

$$\Rightarrow -35 - 7\lambda = 0 \Rightarrow -\lambda = 5 \Rightarrow \lambda = -5$$

33. (b) Obtained means is given by

$$\begin{aligned} \bar{x} &= \frac{0 \cdot 1 + 1 \cdot {}^{50}C_1 + 2 \cdot {}^{50}C_2 + \dots + 50 \cdot {}^{50}C_{50}}{1 + {}^{50}C_1 + {}^{50}C_2 + \dots + {}^{50}C_{50}} \\ &= \frac{\sum_{r=0}^{50} r \cdot {}^{50}C_r}{\sum_{r=0}^{50} {}^{50}C_r} = \frac{\sum_{r=0}^{50} r \cdot \frac{{}^{50}C_{r-1}}{r}}{\sum_{r=0}^{50} {}^{50}C_r} = \frac{50 \cdot 2^{49}}{2^{50}} = \frac{50}{2} \\ \bar{X} &= 25 \end{aligned}$$





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

34. (d) $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$

$$\begin{aligned} \therefore A \cup B &= \{1, 2, 3, 4, 5\} \text{ and } A \cap B = \{3, 4\} \\ A \Delta B &= (A - B) \cup (B - A) = \{1, 2\} \cup \{5\} = \{1, 2, 5\} \\ \therefore n((A \cup B) \times (A \cap B) \times (A \Delta B)) &= 5 \times 2 \times 3 = 30 \end{aligned}$$

35. (b) $n!(21-n)! = 21! \cdot \frac{n!(21-n)!}{21!} = 21! \cdot \frac{1}{{}^{21}C_n}$

For minimum value of $n!(21-n)!$, ${}^{21}C_n$ is maximum.
Maximum value of ${}^{21}C_n = {}^{21}C_{\frac{21-1}{2}} = {}^{21}C_{10}$

$$\therefore \text{Minimum value} = \frac{21!}{{}^{21}C_{10}} = \frac{21!}{21!} \times 10! \cdot 11!$$

Hence, $n = 10$

36. (b) Given that A_i 's are thirty sets with five element each.

$$\text{So, } \sum_{i=1}^{30} n(A_i) = 5 \times 30 = 150 \quad \dots(i)$$

If there are m distinct elements in S and each element of S belongs to exactly 10 of the A_i 's, we have

$$\sum_{i=1}^{30} n(A_i) = 10m \quad \dots(ii)$$

From Eqs. (i) and (ii),

$$10m = 150 \Rightarrow m = 15$$

$$\text{Similarly } \sum_{j=1}^{30} n(B_j) = 3n \text{ and } \sum_{j=1}^{30} n(B_j) = 9m$$

$$\therefore 3n = 9m \Rightarrow n = \frac{9m}{3} = 3m = 3 \times 15 \Rightarrow n = 45$$

37. (c) Given that, equation of ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Coordinate of any point on ellipse with eccentric angle θ are $(a \cos \theta, b \sin \theta)$.

The coordinates of the end point of the latusrectum

$$= \left(ae, \pm \frac{b^2}{a} \right)$$

$$\therefore a \cos \theta = ae \text{ and } b \sin \theta = \pm \frac{b^2}{a}$$

$$\Rightarrow \tan \theta = \pm \frac{b}{ae} \Rightarrow \theta = \tan^{-1} \left(\pm \frac{b}{ae} \right)$$

38. (d) Given that $\alpha^2 = 5\alpha - 3$... (i)

and $\beta = 5\beta - 3$... (ii)

If roots are $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$.

$$\therefore \text{Product of roots} = \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 1$$

$$\text{and sum of root (s)} = \frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{5(\alpha + \beta) - 6}{\alpha\beta} \quad \dots(iii)$$

$\therefore$ Root always satisfies the equation.

From Eq. (i), we can observe the equation whose roots are α and β is

$$x^2 = 5x - 3 \Rightarrow x^2 - 5x + 3 = 0$$

$$\therefore \alpha + \beta = 5 \text{ and } \alpha\beta = 3$$

From Eqs. (iii),

$$S = \frac{5(5) - 6}{3} = \frac{19}{3}$$

Required equation,

$$x^2 - \left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha} \right)x + \frac{\alpha}{\beta} \cdot \frac{\beta}{\alpha} = 0 \Rightarrow x^2 - \left(\frac{19}{3} \right)x + 1 = 0$$

$$\Rightarrow 3x^2 - 19x + 3 = 0$$

39. (d) Given probability that a man who is x years old will die in a year = p

p (no one among $A_1, A_2, A_3, \dots, A_n$ dies within a year)

$$= (1-p)(1-p) \dots n \text{ times}$$

$$= (1-p)^n$$

p (atleast one among $A_1, A_2, \dots, A_n$ dies within a year)

$$= 1 - (1-p)^n$$

p (A_1 dies within a year and is first to die)

$$= \frac{1}{n} [1 - (1-p)^n]$$

40. (d) Given that angle between $\mathbf{a}$ and $\mathbf{b}$ is 120° .

$\therefore \mathbf{a} + x\mathbf{b}, \mathbf{a} - \mathbf{b}$ are right angled.

$$\Rightarrow (\mathbf{a} + x\mathbf{b}) \cdot (\mathbf{a} - \mathbf{b}) = 0$$

$$\Rightarrow |\mathbf{a}|^2 - (x-1)|\mathbf{a}|^2 - 4x|\mathbf{a}|^2 = 0 \quad [\because |\mathbf{b}| = 2|\mathbf{a}|]$$

$$\Rightarrow (2-5x)|\mathbf{a}|^2 = 0 \Rightarrow x = \frac{2}{5} \quad [\because |\mathbf{a}| \neq 0]$$

41. (b) Given condition $x + \frac{100}{x} > 29$, where x is a natural

number. Possible values of x satisfying the above condition are 1, 2, 3, 26, 27, 28, ..., 100.

$$\text{Since, } x + \frac{100}{x} > 29$$

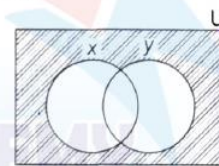
$$x^2 - 29x + 100 > 0$$

$$\Rightarrow (x-4)(x-25) > 0$$

$$\therefore x \in \{1, 2, 3, 26, 27, \dots, 100\} = E \text{ (say)}$$

$$\therefore \text{Probability} = \frac{n(E)}{100} = \frac{78}{100} = \frac{39}{50}$$

42. (c) $X \cap Y' \cap (X \cup Y)'$



$$= X \cap Y' \cap (X' \cap Y') = X \cap ((Y' \cap X') \cap (Y' \cap Y'))$$

$$= X \cap (Y' \cap X') = (X \cap Y') \cap (X \cap X')$$

$$= (X \cap Y') \cap \phi = \phi$$

43. (b) $I = \int e^x (\sinh x + \cosh x) dx$

$$\left\{ \because \frac{d}{dx} (\sinh x) = \cosh x \text{ and } \frac{d}{dx} \cosh x = \sinh x \right.$$

$$\text{and } \int e^x (f(x) + f'(x)) dx = e^x f(x) + C \}$$

$$I = e^x \cosh x + C$$

44. (a) $\mathbf{e}_1 = (1, 1, 1), \mathbf{e}_2 = (1, 1, -1)$

$$\mathbf{e}_1 = 2\mathbf{a} + \mathbf{b} \text{ and } \mathbf{e}_2 = \mathbf{a} + 2\mathbf{b}$$

$$\Rightarrow 2\mathbf{a} + \mathbf{b} = \hat{i} + \hat{j} + \hat{k} \text{ and } \mathbf{a} + 2\mathbf{b} = \hat{i} + \hat{j} - \hat{k}$$

On solving these equations,

$$\mathbf{a} = \frac{1}{3}(\hat{i} + \hat{j} + 3\hat{k}) \text{ and } \mathbf{b} = \frac{1}{3}(\hat{i} + \hat{j} - 3\hat{k})$$





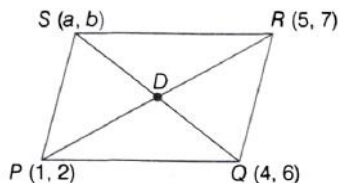
ACME ACADEMY

India's Most Trustable MCA Entrance Academy

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \Rightarrow \cos \theta = \frac{-7}{11}$$

$$\theta = \cos^{-1}\left(\frac{-7}{11}\right)$$

45. (a) Since we know that diagonals of parallelogram bisect each other.



$$D = \left(\frac{a+4}{2}, \frac{b+6}{2} \right) = \left(\frac{1+5}{2}, \frac{2+7}{2} \right)$$

$$\Rightarrow \frac{a+4}{2} = \frac{6}{2} \text{ and } \frac{b+6}{2} = \frac{9}{2}$$

$$\therefore a = 2 \text{ and } b = 3$$

46. (c) $a \cos \theta + b \sin \theta = 2$... (i)

$$a \sin \theta - b \cos \theta = 3$$
 ... (ii)

On squaring Eqs. (i) and (ii) and adding after that

$$(a \cos \theta + b \sin \theta)^2 + (a \sin \theta - b \cos \theta)^2 = 2^2 + 3^2$$

$$\Rightarrow a^2(\cos^2 \theta + \sin^2 \theta) + b^2(\sin^2 \theta + \cos^2 \theta) = 4 + 9$$

$$\therefore a^2 + b^2 = 13$$

47. (d) Since three dice are thrown.

$$\therefore n(S) = 216$$

Triplets are $\{(1, 1, 1), (2, 2, 2), (3, 3, 3), (4, 4, 4), (5, 5, 5), (6, 6, 6)\}$ Number of triplets = 6

$\therefore$ Probability of throwing triplets not more than twice

$$= 1 - \left(\frac{6}{216} \right)^3 = 1 - \left(\frac{1}{36} \right)^3 = 1 - \frac{1}{36^3}$$

48. (d) $I = \int 3^{3^{3^x}} \cdot 3^{3^x} \cdot 3^x dx$

$$\text{Let } 3^{3^{3^x}} = t$$

$$\Rightarrow \frac{d}{dx}(3^{3^{3^x}}) = \frac{dt}{dx} \Rightarrow 3^{3^{3^x}} \cdot 3^{3^x} \cdot 3^x (\log 3)^3 = \frac{dt}{dx}$$

$$I = \frac{1}{(\log 3)^3} \int dt \Rightarrow I = \frac{t}{(\log 3)^3} + C = \frac{3^{3^{3^x}}}{(\log 3)^3} + C$$

49. (a) The number of ways in which a student can attempt one or more questions

$$= {}^{50}C_1 + {}^{50}C_2 + {}^{50}C_3 + \dots + {}^{50}C_{50}$$

$$= 2^{50} - 1 \quad [\because {}^nC_0 + {}^nC_1 + \dots + {}^nC_n = 2^n]$$

50. (b)

Class interval	Frequency	Cumulative frequency
10-20	180	180
20-30	f_1	$180 + f_1$
30-40	34	$214 + f_1$
40-50	180	$394 + f_1$
50-60	136	$530 + f_1$
60-70	f_2	$530 + f_1 + f_2$
70-80	50	$580 + f_1 + f_2$

Given that total, frequency = 686

$$\therefore 580 + f_1 + f_2 = 686 \Rightarrow f_1 + f_2 = 106$$

$\therefore$ Median = 426, so it will lie in 40-50 class interval.

$\therefore$ Modal class = 40 - 50

$\therefore$ Lower limit (l) = 40, h = 10

$$f = 180, cf = 214 + f_1$$

$$\therefore \text{Median} = l + \frac{\left(\frac{N}{2} - cf \right)}{f} \times h$$

$$\Rightarrow 42.6 = 40 + \frac{\frac{686}{2} - (214 + f_1)}{180} \times 10$$

$$\Rightarrow 2.6 \times 18 = 343 - 214 - f_1 \Rightarrow 46.8 = 129 - f_1$$

$$\therefore f_1 = 82.2 \approx 82 \text{ and } f_2 = 106 - 82 = 24$$

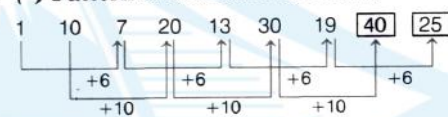
51. (b) VIBGYOR stands for Violet, Indigo, Blue, Green, Yellow, Orange and Red.

So, Yellow, Blue, Red and **Orange** are colours in VIBGYOR.

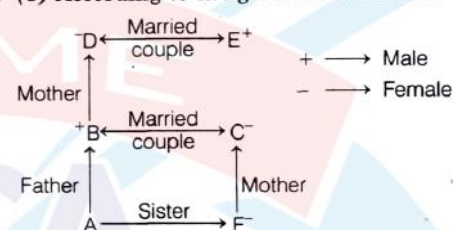
52. (b) Required percentage = $\frac{4817 - 37.16}{37.16} \times 100$

$$= \frac{1101}{37.16} = 29.62\% \approx 31\%$$

53. (*) Pattern of the series as follows



54. (b) According to the given information.



Hence, DE is one of the couples.

55. (a) As, $Z = 52$

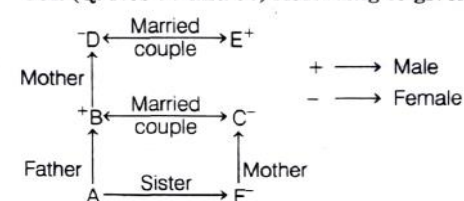
$$\Rightarrow 26 \times 2 = 52$$

and ACT = 48

$$\Rightarrow (1 + 3 + 20) \times 2 = 24 \times 2 = 48$$

$$\text{Similarly, BAT} \Rightarrow (2 + 1 + 20) \times 2 = 23 \times 2 = 46$$

Sol. (Q. Nos 56 and 57) According to given information,



56. (d) None of these statements are true.

57. (c) C is mother of A.

58. (d) 'Trick : Rope : Acrobat' is similar to 'Music: Guitar: Performer'.





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

59. (b) As $9 \times 3 + 8 = 24$

$$\Rightarrow (9 \times 8) \div 3 = 72 \div 3 = 24; 10 \times 2 + 7 = 35$$

$$\Rightarrow (10 \times 7) \div 2 = 70 \div 2 = 35 \text{ and } 80 \times 40 \div 3 = 6$$

$$\Rightarrow (80 \times 3) \div 40 = 240 \div 40 = 6$$

Similarly, $12 \times 4 + 3 = ?$

$$\Rightarrow (12 \times 3) \div 4 = 36 \div 4 = \boxed{9}$$

60. (c) Pattern of series as follows

$$\begin{array}{ccccccc} Q & \xrightarrow{+2} & S & \xrightarrow{+2} & U & \xrightarrow{+2} & W & \xrightarrow{+2} & Y \\ 1 & \xrightarrow{\times 1+1} & 2 & \xrightarrow{\times 2+2} & 6 & \xrightarrow{\times 3+3} & 21 & \xrightarrow{\times 4+4} & 88 \\ F & \xrightarrow{-1} & E & \xrightarrow{-1} & D & \xrightarrow{-1} & C & \xrightarrow{-1} & B \end{array}$$

61. (b) The minimum qualifying marks in Chemistry for eligibility to symposium would lie in the range 20 - 30.

62. (d) According to the question,

$$\begin{array}{l} \triangle \text{ re } \textcircled{\text{me}} \longrightarrow \triangle \text{ he } \textcircled{\text{is}} \text{ late } \dots (i) \\ \text{fa } \textcircled{\text{me}} \text{ la } \longrightarrow \text{she } \textcircled{\text{is}} \text{ larly } \dots (ii) \\ \text{so ti } \triangle \longrightarrow \triangle \text{ he } \text{ leaves soon } \dots (iii) \end{array}$$

From Eqs. (i), (ii) and (iii),
late = re

63. (c) As 7 : 56

$$\Rightarrow 7 : 7 \times (7 + 1) = 7 : 7 \times 8 = 7 : 56$$

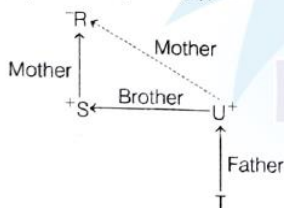
Similarly, 9 : ?

$$\Rightarrow 9 : 9 \times (9 + 1) = 9 : 9 \times 10 = 9 : 90$$

64. (d) Pattern of the series as follows

$$\begin{array}{ccccccc} E & \xrightarrow{+2} & G & \xrightarrow{+2} & I & \xrightarrow{+2} & K & \xrightarrow{+2} & M \\ L & \xrightarrow{+0} & L & \xrightarrow{+0} & L & \xrightarrow{+0} & L & \xrightarrow{+0} & L \\ F & \xrightarrow{+2} & H & \xrightarrow{+2} & J & \xrightarrow{+2} & L & \xrightarrow{+2} & N \\ A & \xrightarrow{+0} & A & \xrightarrow{+0} & A & \xrightarrow{+0} & A & \xrightarrow{+0} & A \end{array}$$

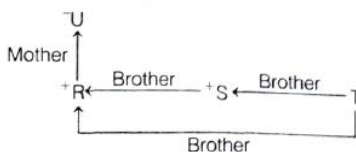
65. (b) From option (a),



$\therefore$ R is grandmother of T.

[incorrect]

From option (b),

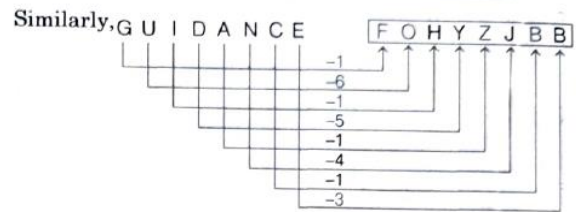
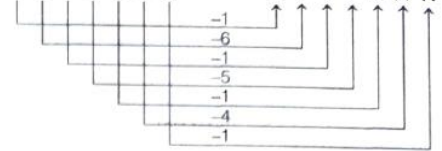


So, option (b) is correct.

66. (d) As, WAITRESS works in RESTAURANT.

Similarly, Teacher works in School.

67. (b) As, C O U N S E L B I T I R A K



68. (d) $15 : 46 \Rightarrow 15 \times 3 + 1 = 46$, $12 : 37 \Rightarrow 12 \times 3 + 1 = 37$,

$$9 : 28 \Rightarrow 9 \times 3 + 1 = 28 \text{ and } \boxed{8 : 33} \Rightarrow 8 \times 3 + 1 = 25 \neq 33$$

69. (a) $\therefore$ Year 2000 is a leap year.

So, if it is Saturday on January 1, 2000, then January 1, 2021 would have been (Saturday + 2 = Monday).

70. (b) 2 yr ago, Rita's age = x (let) and Radha's present age = $2x$

According to the question,

Difference between their ages = 2 yr

$$\Rightarrow 2x - (x + 2) = 2 \Rightarrow 2x - x - 2 = 2 \Rightarrow x = 4$$

Hence, Radha's present age = $4 \times 2 = 8$ yr

71. (c) 22 times in a day are the hands of a clock pointing opposite to each other.

72. (b) Required marks for pursuing studies in Physics

$$= \frac{50 \times 60}{100} = 30$$

Hence, 32 students will be eligible to pursue higher studies in Physics.

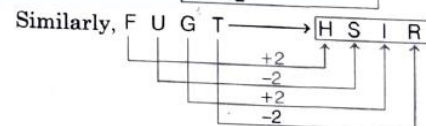
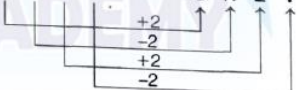
73. (c) 40% of 50 = $50 \times \frac{40}{100} = 20$

Hence, number of students scoring less than 40%.

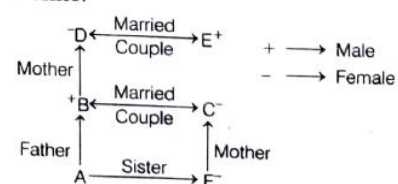
Marks in aggregate = 100 - Number of students scoring 20 and above marks in aggregate

$$= 100 - 73 = 27$$

74. (c) As, B Y C X D W E V



75. (c) According to the given information, Gender of A cannot be defined.



$\therefore$ Number of male members are there in this family cannot be determined.

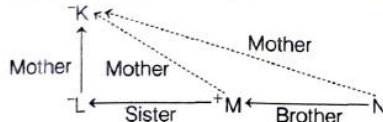




ACME ACADEMY

India's Most Trustable MCA Entrance Academy

76. (a) According to the given information,



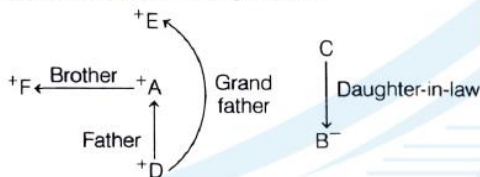
Hence, K is the mother of N.

77. (a) Given,

$$\begin{aligned}
 X &= 1! + 2! + 3! + \dots + 100! \\
 \text{Now, } \frac{X}{24} &= \left(\frac{1! + 2! + 3! + \dots + 100!}{24} \right) \\
 \Rightarrow \frac{X}{24} &= \left(\frac{1 + 2 \times 1 + 3 \times 2 \times 1 + \dots + 100 \times 99 \times \dots \times 1}{24} \right) \\
 \Rightarrow \frac{X}{24} &= \left(\frac{1 + 2 + 6 + 24 + 120 + \dots + 100 \times 99 \times \dots \times 1}{24} \right) \\
 \Rightarrow \frac{X}{24} &= \frac{9}{24}
 \end{aligned}$$

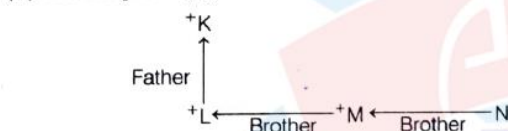
$\therefore$ Required remainder = 9

78. (d) According to the question,



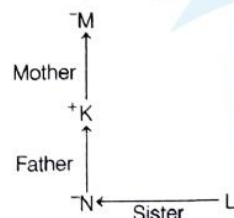
Hence, relationship of C with F cannot be defined.
Hence, data inadequate.

79. (b) From option (a),



$\therefore$ Gender of N cannot be defined. (Incorrect)

From option (b),



Hence, N is definitely daughter of K. (correct)

80. (b) Crop is a necessary part of Harvest.
81. (d) Required difference = Number of students scoring 30 and above marks in Chemistry - Number of students scoring 30 and above marks in aggregate = 27 - 21 = 6

82. (c) Rate of increase for Product

$$A = \frac{4817 - 2015}{2015} \times 100\% = 139.05\%$$

Rate of increase for Product

$$B = \frac{37.16 - 5.93}{5.93} \times 100\% = 526.64\%$$

Rate of increase for Product

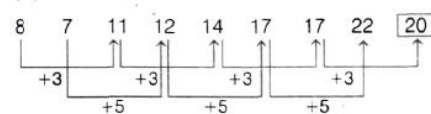
$$D = \frac{12.21 - 7.88}{7.88} \times 100\% = 54.94\%$$

Rate of increase for Product

$$E = \frac{10.19 - 5.01}{5.01} \times 100\% = 103.39\%$$

Hence, during the period 2005-2010, the minimum rate of increase in sales is in the case of product D.

83. (b) Pattern of the series as follows



84. (c) As, Pulsate and Throb are synonyms.
Similarly, Examine and Scrutinize are synonyms.

85. (d) Total number of students in the class = 50

Let Paul's rank = x

So, Raghu's rank = 2x

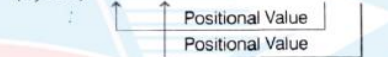
$\therefore$ 10 students who have ranked worse than that of Raghu.

$\therefore$ Raghu's rank = 40

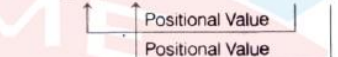
So, Paul's rank = $\frac{40}{2} = 20$ th

86. (c) Except Hyena, all others are vegetarian.

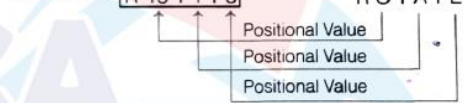
87. (b) As, A3T15R



and D1T5



Similarly, R15T1T5



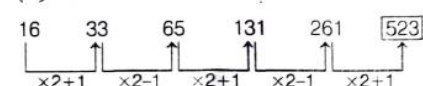
88. (d) Rate of increase for product

$$C = \frac{2914 - 1497}{1497} \times 100 = 94.65\%$$

and from answer (82). It is clear, sales have increased by nearly 55% from 2005 to 2010 in case of product D.

89. (c) Except beak, all others are parts of an airplane.

90. (a) Pattern of the series as follows



91. (b) Pipe-lining is the process of improving the performance of the system by processing different instructions at the same time, with only one instruction performing one specific operation.

92. (d) Accumulators are used by ALU (Arithmetic Logic Unit) to store the intermediate results. The ALU is the computational center of the CPU. It performs all mathematical and logical operations.





ACME ACADEMY

India's Most Trustable MCA Entrance Academy

93. (d) 1 GigaByte (GB) = $1024 = 10^{10}$ MB
1 TeraByte (TB) = 2^{10} GB
1 PetaByte (PB) = 2^{10} TB
1 ExaByte (EB) = 2^{10} PB
 $= 2^{10} \times 2^{10}$ TB
 $= 2^{10} \times 2^{10} \times 2^{10}$ GB = 2^{30} GB

94. (b) The cache memory is more effective because of locality of reference. This means that the cache depends on the location in the memory that is referenced often.

95. (a) Registers are fastest means of access for CPU. Registers are the small memory locations which are present closet to the CPU.

96. (c) Binary equivalent of $(2217)_8$ is

$$\begin{array}{cccc} 2 & 2 & 1 & 7 \\ \downarrow & \downarrow & \downarrow & \downarrow \\ 010 & 010 & 001 & 111 \\ = (010010001111)_2 \end{array}$$

Hexademical equivalent of $(010010001111)_2$ is

$$\begin{array}{ccc} 0100 & 1000 & 1111 \\ \downarrow & \downarrow & \downarrow \\ 4 & 8 & 15 \rightarrow F = (48F)_{16} \end{array}$$

So, $(2217)_8 = (048F)_{16}$

97. (a) To fetch data from the secondary memory, MAR (Memory Address Register) is used. MAR is a CPU register that either stores the memory address from which data will be fetched to the CPU or the address to which data will be sent and stored.

98. (b) Binary multiplication for 00×11

$$\begin{array}{r} 00 \\ \times 11 \\ \hline 00 \\ 00 \times \\ \hline 000 \end{array}$$

So, it will give output as 00.

99. (d) Number of steps = $2n^2$

Speed of computer = 10^6 per second

For $n = 10000(10^4)$

$$\begin{aligned} \therefore \text{Time} &= \frac{\text{Number of steps}}{\text{Speed of computer}} \\ &= \frac{2n^2}{10^6} = \frac{2 \times (10^4)^2}{10^6} = \frac{2 \times 10^8}{10^6} \\ &= 2 \times 10^2 = 200 \text{ sec} \end{aligned}$$

100. (a) To access the I/O devices, the status flag is continuously checked in program controlled I/O. In this method, the processor constantly checks the status flags and when it finds that the flag is set it performs the appropriate operation.

101. (b) Mere knowledge and culture may be distinguished from each other in that the former gratifies curiosity, the latter refines personality.

102. (b) According to the Ancient Indian Philosophers, Education brings about salvation and development of the soul.

103. (c) Scintillating means sparkling or shining brightly. So, glittering is the most suitable synonym of the given word. Glittering also means shining with a shimmering or sparkling light.

104. (d) Bland means unseasoned, mild-tasting or insipid (of food or drink). So, 'tasteless' is the most suitable synonym of the given word.

105. (c) 'Lead weights' means hindering factors, of pride and prejudice, passion, and desire, in the liberation of the soul.

106. (c) The function of university is to prepare the young to take their place in society.

107. (a) According to the passage, the function of the teacher is to remove leaden weights of pride and prejudice, passion and desire to liberate the soul.

108. (b) Passion and desire, pride and prejudice are the hindering factors in the liberation and development of the soul.

109. (c) According to the given context, 'explain' is the most appropriate word for the given blank.

110. (c) Recuperate means recover from illness or exertion. Hence, 'recover' is its correct synonym.

111. (c) As per the given context, 'ease' is the most suitable word for the given blank.

112. (d) According to Goethe, the object of education is to form taste.

113. (a) The man's culture judged by the quality of mind which brings to bear on the facts of life.

114. (c) Erudite means having or showing great knowledge or learning. So, among the given, illiterate is the most suitable antonym of the given word.

115. (a) Barbarian is a person who considered extremely primitive and uncivilised.

116. (a) Impounded is the correct word for the given blank. Impounded means seize and take legal custody of something (especially a vehicle, goods or documents).

117. (b) 'Incurred' is the most suitable word for the given blank. Incurred means to experience something unwelcome or unpleasant.

118. (b) The most suitable antonym of given word 'Negligence' is 'Meticulousness'.

Negligence means failure to take proper care over something.

Meticulousness means showing extreme or excessive care in thinking about or dealing with small details a meticulous description.

119. (c) One who does not save for future is called improvident.

120. (d) 'High' is the appropriate word for the given blank.

